

Remarks on Bedert’s “On unique sums in Abelian groups”

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This note collects corrections, sharpenings, finite-group results, and data related to Bedert’s paper [B]. Its controlling purpose is verification: every item below was produced while checking, extending, or reproducing [B]’s argument, and each is presented with its own verification tier rather than as one combined new result. Parts II and III contain observations obtained in the course of checking the argument. They are offered as supplements to the paper, not as claims about what its author intended to optimize. For a finite subset A of an Abelian group, [B] calls $a_1 + a_2$ a unique sum when its only ordered representations are (a_1, a_2) and (a_2, a_1) .

For $A \subset \mathbb{Z}/p\mathbb{Z}$, Scheinerman [S, Definitions 3.1–3.2] had earlier treated the same condition and, writing $\nu(A)$ for the minimum ordered representation count on $A + A$, stated that A is unique-sum-free exactly when $\nu(A) \geq 3$. Green records that Andrew Granville (personal communication) and Swastik Kopparty (Harvard 2017 open problems session) each asked him this prime-cyclic minimum question [G, Problem 27].

Part I

Corrections and scope

1 A 2-torsion scope issue

At [B, line 291], the proof says that a set with no unique sum is balanced: the sum $a + a$ has another representation, which is taken to have distinct summands. In a group with 2-torsion, however, the other representation may be another diagonal pair. Consider

$$A = \{0, 1, 2, 3, 5, 6, 8\} \subset \mathbb{Z}/12\mathbb{Z}.$$

The numbers of unordered representations of the twelve residues are

$$(2, 2, 3, 2, 3, 2, 3, 2, 3, 2, 2, 2).$$

Thus A has no unique sum under [B, line 62]. It is not balanced: neither 0 nor 6 is the midpoint of two distinct members of A . The proof of the general-group lower-bound theorem invokes “no unique sum implies balanced” at [B, line 458]. The printed proof therefore needs an odd-order clause at that step, or a separate argument for groups with 2-torsion. This does not dispute the theorem itself.

The boundary is sharp among cyclic groups of even order.

Theorem. *For $m \geq 0$, there is a set $A \subset \mathbb{Z}/(2m)\mathbb{Z}$ with no unique sum and some ordered representation count equal to 2 if and only if $m \geq 6$.*

For $m \geq 6$, one may take

$$\{0, \dots, m\} \setminus \{m - 2\} \cup \{m + 2\}.$$

VERIFICATION: KERNEL-CERTIFIED. The family and the exhaustive cases $2m < 12$ are checked in `CycEven.lean`; `verify.py` recomputes the C_{12} table.

2 The Luczak–Schoen upper bound

Luczak and Schoen [LS, Theorem 2] use the ordered representation count of [B]. Taking $Q = 2$ gives, for $p \geq 2^{22}$, a set $A \subset \mathbb{Z}/p\mathbb{Z}$ with

$$|A| \leq 4(\log_2 p)^2, \quad r_A(g) \geq 4 \quad (g \in A + A).$$

Hence $m(p) \leq 4(\log_2 p)^2$. Thus the order of growth in [B, Theorem 5] already follows from [LS]. Bedert’s construction improves the constant and is more direct. The author has confirmed this observation.

VERIFICATION: SOURCE-GROUNDED CONSEQUENCE OF PUBLISHED THEOREMS.

3 A redundant factor in Proposition 1

For an indexed family $Z = (z_1, \dots, z_n)$, let $\Sigma(Z)$ be its set of subset sums and let d be the largest size of a dissociated index set. Then

$$|\Sigma(Z)| \leq \#\{I \subseteq [n] : I \text{ is dissociated}\} \leq \sum_{j=0}^d \binom{n}{j} \leq \binom{n+d}{d}.$$

Choose the colexicographically least index set representing each subset sum. Every set shattered by this family is dissociated, and Pajor’s strengthening of Sauer–Shelah completes the count. This removes the first factor from $\binom{n}{d} \binom{n+d}{d}$ in [B, Proposition 1].

VERIFICATION: KERNEL-CERTIFIED FIRST TWO INEQUALITIES; FINITE ARITHMETIC CHECK FOR THE LAST. `SubsetSums.lean` checks the first two inequalities and `verify.py` checks the Vandermonde step. The author has explained that he chose the simpler sufficient bound because the factor does not affect the asymptotic theorem.

4 The empty set

The definition at [B, line 282] makes the empty set balanced, while Proposition 3 asks for $g \in -B$ for every balanced B . The repair is to require balanced sets to be nonempty in the definition, or in the propositions that use such an element.

VERIFICATION: SOURCE-GROUNDED DEFINITIONAL CORRECTION.

Part II

The general-group question

5 The minima three and four

Theorem. *For every finite Abelian group G ,*

$$m(G) = 3 \iff 3 \mid |G|.$$

Moreover, if $4 \mid |G|$ and $3 \nmid |G|$, then $m(G) = 4$.

Proof. Translate a three-set to $\{0, x, y\}$. If it has no unique sum, the pair $\{0, x\}$ can collide only with $\{y, y\}$, so $x = 2y$. Symmetrically $y = 2x$, whence $3x = 0$. Conversely, a coset of a subgroup of order three has no unique sum. This proves the first assertion by Cauchy’s theorem.

If $4 \mid |G|$, choose a subgroup H of order four. For $a, b \in H$, choose $h \in H \setminus \{0, b - a\}$. Then

$$(a + h) + (b - h) = a + b$$

is neither the original nor the swapped pair. Thus H has no unique sum. The first assertion and the elementary two-point obstruction give the lower bound when $3 \nmid |G|$. $\square$

VERIFICATION: PROVED, ELEMENTARY; ALGEBRAIC CORE KERNEL-CERTIFIED. `Dcyc.lean` checks the three-set algebra, both order-four groups, and the necessary C_{12} counterexample to the unguarded four-branch.

6 Census-52

VERIFICATION: EXACT COMPUTATION, DRAT-CERTIFIED. All 88 Abelian groups of orders 2 through 52, and 26 separately labelled stress groups through order 221, match

$$m(G) = \min(\{4 : 4 \mid |G|\} \cup \{m(p) : p \text{ odd prime, } p \mid |G|\}), \quad (2)$$

with C_2 separately having no nonempty such set. This is 114 exact groups, not a theorem for all finite Abelian groups. Each of the 114 lower bounds has a retained DRAT proof and verified checker log; the 113 upper witnesses were recounted directly. The proofs are not distributed with this note. Their CNF and proof SHA-256 hashes, paths, group sizes, and outcomes are recorded in `mg-census-through-52.json`, whose own SHA-256 is

f8d1c7119e5e98f6d97c7bd4bc76e7eeac1060915876196040f6df74f9a073e9.

The manifest does not record proof byte counts, so no byte-size claim is made here. This census tests and extends to noncyclic groups the cyclic conjecture stated by Scheinerman [S, Conjecture 3.40].

7 A double-cyclic conjecture

Conjecture. *For every odd prime p ,*

$$m(C_{2p}) = m(C_p). \quad (3)$$

This is the $n = 2p$ case of Scheinerman’s published conjecture $\mu(n) = \min_{q|n, q \text{ prime}} \mu(q)$ [S, Conjecture 3.40]. The following evidence and reduction do not prove (3).

First, a DRAT-certified exact search shows that every minimum set in $C_2 \times C_p$ lies in one parity fibre for $p = 3, 5, 7, 11, 13, 17, 19, 23, 29$. Second, write

$$A_e = \{x : (e, x) \in A\}, \quad B = A_0 \cup A_1, \quad D = A_0 \cap A_1.$$

If A is minimum and $D = \emptyset$, projection is injective and B has no unique sum. Hence $m(C_p) \leq |B| = |A|$, while embedding a minimum set of C_p in one fibre gives the reverse inequality.

VERIFICATION: PROVED REDUCTION. Thus empty overlap at the minimum implies (3). Eliminating nonempty residual overlap remains open.

The naive statement that projection preserves the property is false. In $C_2 \times C_{11}$, the full double lift of

$$\{0, 1, 2, 4, 7\} \subset C_{11}$$

has no unique sum, although its projection does. This counterexample is kernel-certified in `Dcyc.lean`. It is not minimum, so it does not refute (3). It is affinely equivalent to Scheinerman’s example [S, Section 3.6].

Part III

Sharpenings of the prime-cyclic argument

For comparison, if $\alpha(p)$ denotes the minimum cardinality of a balanced subset of $\mathbb{Z}/p\mathbb{Z}$, Nedev proved

$$\alpha(p) \geq \log_2 p + 3 - \log_2 3$$

for every odd prime p [N, Corollary 9]. Every unique-sum-free set in $\mathbb{Z}/p\mathbb{Z}$ is balanced, and therefore

$$m(p) \geq \alpha(p) \geq \log_2 p + 3 - \log_2 3.$$

This transfer uses only the inclusion of the unique-sum-free sets among the balanced sets; it does not identify the two problems. Balancedness constrains the diagonal sums $2x$, whereas the unique-sum-free condition constrains every sum in $A + A$. The transferred bound is weaker than the asymptotic lower bound in [B].

VERIFICATION: PUBLISHED THEOREM PLUS ELEMENTARY TRANSFER.

8 A registered exponent-two lower bound

Palomar Registry entry PALOMAR-2026-08-18-000003 v1 records Xinjie He’s Lean formalization of a theorem from his own working draft. For all sufficiently large primes p , it proves

$$m(p) \geq c \left(\frac{\log p}{\log \log p} \right)^2$$

for an absolute constant $c > 0$. Together with the published upper bound $m(p) \ll (\log p)^2$, this determines the exponent of $\log p$ up to lower-order factors.

VERIFICATION: KERNEL-REPLAYED ON OUR HARDWARE; REPLAY LOG AND AXIOM PRINTS IN THE README. This tier applies to the formal theorem and not to a line-by-line referee review of the proof. The Palomar entry is a registered formal artifact, not a peer-reviewed publication. Its source describes the mathematics as the author’s working draft; no separate preprint registration is asserted here. The immutable entry is PALOMAR-2026-08-18-000003 v1.

9 The subsequent Cao–Yuan preprint

Cao and Yuan [CY] put $G_p := \mathbb{Z}/p\mathbb{Z}$. For a finite Abelian group G with $|G| \geq 3$, they define $m(G)$ as the minimum cardinality of a subset $A \subseteq G$, with $|A| \geq 2$, having no unique sum. Their main theorem states:

Theorem (Cao–Yuan, preprint). *For every prime p , every set $A \subseteq G_p$, with $|A| \geq 2$ and no unique sum, satisfies*

$$|A| \geq c_* \log p \log \log p.$$

Consequently,

$$m(p) \gg \log p \log \log p.$$

Here $c_* = 1/40000000010006048640$ in their explicit statement. The source is arXiv:2608.06728v1, submitted 7 August 2026, and is a preprint. Its source reports a Lean 4 check, but the arXiv source bundle contains no Lean artifact; that check has not been independently kernel-verified here.

Their exact theorem and the exact theorem below are overlapping but incomparable. The former bounds $|A|$ and uses a level-four additive dimension in its structural proposition; the latter bounds Bedert’s $K(A) = |A|/\dim A$ with a particular constant and additional displayed hypotheses. Neither exact statement implies the other as stated. For the derived asymptotic role in bounding $m(p)$, however, the Cao–Yuan preprint’s $\log p \log \log p$ lower bound strictly dominates the bound derived from the theorem below. No theorem stated in this note is refuted.

VERIFICATION: CAO–YUAN: UNREFEREED PREPRINT; COMPARISON: SOURCE-GROUNDED ADJUDICATION.

10 A one-log improvement

Put

$$M = \log_2 \log_2 p, \quad K(A) = \frac{|A|}{\dim A}.$$

Replacing the update $|S'| \leq \max(2|S|, |S|^3)$ in [B, lines 479–494] by a two-point update throughout gives the following conclusion under the hypotheses displayed in the formal statement:

$$K(A) > \sqrt{M/404}. \tag{1}$$

This is a genuine asymptotic improvement, replacing the triple logarithm under the square root by a double logarithm. The free inequality $K(A) \geq 1$ dominates (1) whenever

$$p < 2^{2^{404}}.$$

There are two formal statements. The statement in `ObjectLayer/07Headline.lean` is

VERIFICATION: CONDITIONAL, KERNEL-CERTIFIED; FRONTIER HYPOTHESIS. conditional on the stated Lev rectification interface `hBLR`. In that formalization we do not establish this hypothesis, and nothing in the development supplies an instance of it. The extracted theorem `sqrt_improvement_from_isUSF_internal` has the identical remaining hypotheses and conclusion but no `hBLR` binder.

VERIFICATION: KERNEL-CERTIFIED PER THE RECORDED ORIGINAL ELABORATION; “UNCONDITIONAL” ONLY IN THE ABSENCE OF THE `hBLR` BINDER; NOT FRESHLY RE-VERIFIABLE ON THE PRESENT 16 GiB HOST; IMPORT CLOSURE AND AXIOM PRINT IN THE README. The associated satisfiability check supplies a concrete model of the arithmetic hypotheses, so the unconditional statement is not vacuous merely by inconsistent numeric assumptions. The paper itself says that the simplest sufficient form of this slowly growing factor was chosen and that its exact shape was not optimized [B, after Theorem 3]. As a contribution to the asymptotic lower bound for $m(p)$, this result is superseded in role by both later bounds in this note: the Palomar theorem above already gives the strongest asymptotic exponent, and the Cao–Yuan preprint [CY] gives an intermediate one. Its exact theorem, formal artifact, and verification tier are unchanged.

11 Sharpness of the simultaneous-selector threshold

Let $f(p)$ be the largest cardinality for which the selector conclusion used in [B, Lemma 5.1] holds for every $S \subset \mathbb{Z}/p\mathbb{Z}$. Then

$$f(p) = \lceil \log_2 p \rceil.$$

Equivalently, a selector exists for every k -set precisely while $2^{k-1} < p$. The general upper obstruction is

$$\{0\} \cup \{ \lceil p/2^i \rceil : 1 \leq i \leq \lceil \log_2 p \rceil \}.$$

VERIFICATION: EXHAUSTIVE COMPUTATION AT 29 PRIMES ACROSS FIVE COMPLETE BANDS, PLUS A HAND PROOF OF THE GENERAL UPPER BOUND WITH MACHINE-CHECKED ALGEBRAIC IDENTITIES; NOT KERNEL-CERTIFIED. The exact computations cover all primes in the five bands from (4, 8) through (64, 128). In particular, all 234,531,275 seven-subsets pass at $p = 127$, while an eight-subset fails. The rectification threshold used here originates in Lev’s Theorem 1 [L]. The gain over the displayed $2^{|S|} \leq p$ hypothesis is exactly one element, not a change of scale.

Part IV

Data

12 Exact values of $m(p)$

p	3	5	7	11	13	17	19	23	29	31	37	41	43	47	53	59
$m(p)$	3	4	5	7	7	8	9	10	11	11	12	13	13	13	14	15

p	verification tier
3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37	exact solver output, independently corroborated; not certified here
41, 43, 47	DRAT-certified optimality and verified witness
53, 59	solver-certified optimality and verified witness: per-cube proofs checked by a verified checker, hash-bound LRAT retained, raw DRAT generated and checked at generation time but not retained

The same sixteen values were printed earlier in Scheinerman’s Table 3.1 [S]. That source gives witnesses but does not state an optimality-verification method; the tiers in this table describe the evidence retained here. The checking script redoes the small cases feasible in pure Python and verifies the displayed witnesses at 41, 43, 47, 53, 59.

A parse-level byte-tamper theorem now closes one part of the search machinery’s own encoding-to-DIMACS-byte trust boundary (audited in `statement-audit.md`): a single flipped literal in the production DIMACS bytes is caught by parsing rather than silently accepted. Its exact tier is kernel-checked-on-cloud: checked on a documented cloud host, not yet independently replayed on this note’s own local machine, so it is stated at that tier and not as kernel-certified. Direct elaboration of the target statement measured 32 seconds at a peak of about 8.0 GiB; an earlier apparent reading near 40 GiB was later attributed to a one-time build of a separate dependency, not to the target theorem’s own cost.

Checking

Run

```
lake exe cache get && lake build    and    python3 verify.py.
```

The principal Lean axiom prints should contain exactly `propext`, `Classical.choice`, and `Quot.sound`. This command freshly rebuilds every module in the note except the internal one-log theorem’s closure (“A one-log improvement” above), which is included for inspection but, per its stated tier there, is not freshly re-verifiable on a 16 GiB host; see the README for that module’s import-closure record and axiom print.

AI tool disclosure

The mathematics in this note, including its proofs, counterexamples, and verification code, was produced in substantial part by AI tools (Claude and OpenAI Codex models) working under this note’s author’s direction. Trust in the results

therefore rests not on how they were drafted but on the verification artifacts attached to each claim. Kernel-certified statements are checked by the Lean kernel, and where stated, independently re-checked by a second kernel pass; computational claims carry retained certificates, including DRAT certificates checked by an independent checker; and quoted results were verified against the cited sources. This note’s author directed the work, made every judgment call about what to claim and at what verification tier, and retains sole responsibility for its correctness.

References

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